

Levinson's method in short intervals and simple zeros of the zeta function

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CAOS Research program

github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Fix $1/2 < \theta < 1$. We localize Levinson's method, with Conrey's general operator polynomial Q , to the intervals $(T, T + T^\theta]$ for mollifiers of length T^ν with $\nu < \theta - 1/2$, by running Young's short proof of the mollified second moment with a weight supported on the window. We then show that on such windows the method counts distinct sign changes of Hardy's function, that is, distinct critical zeros of odd order, with lower density $\kappa = 1 - R^{-1} \log c(P, Q, R, \nu)$ for every Q with $Q(x) + Q(1 - x)$ constant. For operator polynomials of degree 201 we certify $\kappa > 0.7170\nu$ for ν between 0.0199 and 0.0999. Used as the odd-support input of the Hilbert-parity inequality $(Q - S)(N - O) \geq 2(N - S)^2$, together with Wang's short-interval pair-correlation theorem, this gives a positive proportion of simple zeros on the critical line in $(T, T + T^\theta]$ for every fixed $\theta \in [0.534, 1)$. The previous exponents for this statement were 0.552 (Steuding), 0.55019... (Wang) and 0.5458838 (the same parity inequality with a rank-six Selberg detector). At $\theta = 0.5459$ the lower density exceeds 0.017763, about one thousand times the previous bound, and numerically the new bound improves the previous ones for θ below about 0.567. All constants are certified with exact rational and ball arithmetic and replayed independently by validated quadrature. The results are asymptotic, use Wang's recent preprint for the pair term, and do not prove the Riemann hypothesis.

1 Introduction

Let $\rho = \beta + i\gamma$ denote the nontrivial zeros of the Riemann zeta function. For $T, H > 0$ let $N(T, H)$ count the zeros with $T < \gamma \leq T + H$, with multiplicity; let $S(T, H)$ count the simple zeros among them with $\beta = 1/2$; and let $O(T, H)$ count the distinct points $t \in (T, T + H]$ at which Hardy's function $Z(t) = e^{i\vartheta(t)}\zeta(1/2 + it)$ changes sign, that is, the distinct critical zeros of odd multiplicity. Throughout, $H = T^\theta$ with a fixed $\theta \in (1/2, 1)$ and $L = \log T$.

Selberg proved that $O(T, T^\theta) \gg N(T, T^\theta)$ for every $\theta > 1/2$, and Karatsuba extended this to $\theta > 27/82$ [5]. These theorems do not give simple zeros. For simple critical zeros, Steuding

localized Levinson's method with a linear operator polynomial, proved a short-window mollified moment with error $O(T^{1/3+\epsilon}M^{4/3})$, and obtained a positive proportion for $\theta \geq 0.552$ [8, 9]; he also observed that the mollifier range available from the classical off-diagonal treatment gives only $\theta \geq 0.693$ for linear Q . Wang's short-interval pair-correlation theorem [10] gives the lower density

$$c(\theta) = 2 - \frac{\theta}{2} - \frac{1}{\sqrt{2}} \cot \frac{\theta}{\sqrt{2}}, \quad (1)$$

which is positive for $\theta > \theta_0 = 0.550193964744\dots$. The finite Hilbert–parity inequality of [2, Proposition 2.5] converts any lower density k for O/N into the simple-critical density

$$h(\theta; k) = \frac{3 + k - \sqrt{(1-k)(9-k-8c(\theta))}}{4}. \quad (2)$$

With the rank-six Selberg detector of Pearce-Crump [6], $k_6(\theta) = (\theta - 1/2)/(4eC_6)$ with $C_6 = 0.65663\dots$, the resulting term is positive exactly for $\theta > \theta_6$, where $0.5458837 < \theta_6 < 0.5458838$ [2, Theorem 1.2]. Since the positivity condition is $k > 1 - 2/(2 - c(\theta))$, the onset is governed by the size of the odd-support density near $\theta = 1/2$, where k_6 has slope only $1/(4eC_6) \approx 0.140$.

This paper replaces the Selberg detector by Levinson's. The mollified second moment behind Levinson's method is available in short windows (Theorem 1.2); the localized method counts distinct sign changes of Z for Conrey's operator polynomials of every degree (Theorem 1.3); and polynomials of high degree push the density to about $0.717(\theta - 1/2)$, five times the Selberg slope. The labels [D] and [MV] mark deductions and finite machine-verified statements.

Theorem 1.1 (Simple critical zeros from $T^{0.534}$, D+MV). *For every fixed $\theta \in [0.534, 1)$,*

$$\liminf_{T \rightarrow \infty} \frac{S(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2\kappa_\theta}{3}, h(\theta; \kappa_\theta) \right\} > 0, \quad (3)$$

where κ_θ is any certified value of Table 1 with $\nu < \theta - 1/2$. In particular the right side is at least $h(\theta; \kappa_{0.0339})$ with $\kappa_{0.0339} > 0.0243176419$, which is positive and increasing on $[0.534, 1)$, and with $\nu = \theta - 1/2 - 10^{-4}$

θ	0.534	0.535	0.54	0.5459	0.55
$h(\theta; \kappa_\theta) >$	$1.4806 \cdot 10^{-5}$	0.0015246	0.0090231	0.0177638	0.0237708

At every point of the table $c(\theta) < 0$ and $h(\theta; \kappa_\theta)$ is the largest term. Numerically, h with $\kappa \approx 0.7173(\theta - 1/2)$ stays above $c(\theta)$ up to $\theta \approx 0.567$; beyond that point Wang's term $c(\theta)$ is the largest in (3). The two analytic inputs are the following.

Theorem 1.2 (Mollified moment on a short window, D). *Let $0 < \nu < \theta - 1/2$, $R > 0$, let P be a real polynomial with $P(0) = 0$, $P(1) = 1$, and let Q be a real polynomial with $Q(0) = 1$. Put $\sigma_0 = 1/2 - R/L$, $y = T^\nu$, $\Delta = H/L$,*

$$V(s) = Q\left(-\frac{1}{L} \frac{d}{ds}\right) \zeta(s), \quad \psi(s) = \sum_{h \leq y} \frac{\mu(h)}{h^{s+1/2-\sigma_0}} P\left(\frac{\log(y/h)}{\log y}\right).$$

Let w be smooth with $0 \leq w \leq 1$, $w = 1$ on $[T, T+H]$, support in $[T-\Delta, T+H+\Delta]$ and $w^{(j)} \ll_j \Delta^{-j}$. Then

$$\int_{\mathbb{R}} w(t) |V\psi(\sigma_0 + it)|^2 dt = c(P, Q, R, \nu) \hat{w}(0) + O(H/L),$$

$$c(P, Q, R, \nu) = 1 + \frac{1}{\nu} \int_0^1 \int_0^1 (w_R(v) P'(u) + \nu w'_R(v) P(u))^2 du dv, \quad w_R(v) = e^{Rv} Q(v).$$

Theorem 1.3 (Distinct sign changes, D). *In the setting of Theorem 1.2, suppose that $Q(x) + Q(1-x)$ is a nonzero constant. Then*

$$\liminf_{T \rightarrow \infty} \frac{O(T, T^\theta)}{N(T, T^\theta)} \geq \kappa(P, Q, R, \nu) := 1 - \frac{1}{R} \log c(P, Q, R, \nu).$$

Theorem 1.2 is Conrey's constant [4] in the form given by Young [11], on a window of length T^θ . The only place where the window length restricts the mollifier is the off-diagonal term, which forces $\nu < \theta - 1/2$, the analogue of Levinson's range $\nu < 1/2$ for $\theta = 1$. Theorem 1.3 is a sign-change sharpening of the classical counting argument: zeros of the Levinson function on the critical line carry full weight, and the argument of the detector is followed through its level crossings. The classical sources conclude simple zeros only for $\deg Q = 1$ [4, 1]; for higher degree the method does not separate simple zeros from zeros of odd order at least three, and that is exactly the information the parity inequality requires.

We could not locate a statement of Theorem 1.2 for general Q on windows shorter than T , or of Theorem 1.3 for $\deg Q \geq 3$; Conrey's Gaussian-window proposition [4] applies to windows $T^{1-\delta}$ with $\delta < 1/7 - \nu/4$. The search was bounded and is not a priority determination.

2 The mollified moment on a short window

We follow Young [11]. His Theorem 2 treats a weight supported in $[T/4, 2T]$ with $w^{(j)} \ll (T/L)^{-j}$ and a mollifier of length $M = T^\vartheta$, $\vartheta < 1/2$. Here the weight has support of length at most $2H$, the derivative scale is $\Delta = H/L$, and M is replaced by $y = T^\nu$. We list every step in which the length enters; all others are used verbatim. As in Young's Section 3, it suffices to prove

$$I(\alpha, \beta) := \int w(t) \zeta(\tfrac{1}{2} + \alpha + it) \zeta(\tfrac{1}{2} + \beta - it) |\psi(\sigma_0 + it)|^2 dt = c(\alpha, \beta) \hat{w}(0) + O(H/L) \quad (4)$$

uniformly for $\alpha, \beta \ll 1/L$, with $c(\alpha, \beta)$ Young's (3.3) with ϑ replaced by ν and M by y . We first work on the annuli $\alpha, \beta \asymp 1/L$, $|\alpha + \beta| \gg 1/L$ of Young's Lemma 6. There the factor $G(s) = e^{s^2} p(s)$, $p(s) = ((\alpha + \beta)^2 - 4s^2)/(\alpha + \beta)^2$, of Young's approximate functional equation (Lemma 4) satisfies $p(s) \ll L^2(1 + |s|^2)$. Hence the weight $V_{\alpha, \beta}(x, t)$ and the factor $X_{\alpha, \beta, t}$ of that lemma obey, for $t \geq T/2$,

$$t^j \partial_t^j V_{\alpha, \beta}(x, t) \ll_{A, j} L^2 (1 + x/t)^{-A}, \quad X_{\alpha, \beta, t} = (t/2\pi)^{-\alpha-\beta} (1 + O(t^{-1})). \quad (5)$$

The first bound is Young's (4.3), whose decay factor is printed as $(1 + |t/x|)^{-A}$; the Mellin representation (4.1) gives $(1 + x/t)^{-A}$, which is the form used in his proof of Lemma 5. The Stirling expansion (4.2) of $g_{\alpha, \beta}(s, t)$ is uniform only for $|s| = o(t^{1/2})$; on every vertical line used below, the part $|\Im s| \geq T^{1/3}$ is negligible because $|G(s)| \ll L^2 e^{-(\Im s)^2/2}$ there, while $|g_{\alpha, \beta}(s, t)| \ll (t + |s|)^{O(1)}$.

Lemma 2.1 (Twisted integral). *Let $h, k \leq y$ and let α, β lie on the annuli. Then*

$$\begin{aligned} \int w(t) \left(\frac{h}{k}\right)^{-it} \zeta(\tfrac{1}{2} + \alpha + it) \zeta(\tfrac{1}{2} + \beta - it) dt &= \sum_{hm=kn} \frac{\int V_{\alpha, \beta}(mn, t) w(t) dt}{m^{1/2+\alpha} n^{1/2+\beta}} \\ &+ \sum_{hm=kn} \frac{\int V_{-\beta, -\alpha}(mn, t) X_{\alpha, \beta, t} w(t) dt}{m^{1/2-\beta} n^{1/2-\alpha}} + O_A(T^{-A}). \end{aligned}$$

Proof. Insert the approximate functional equation. Its error term contributes $O(HT^{-A})$ and the terms with $hm = kn$ are the main terms. For $hm \neq kn$, the bounds (5) and $w^{(j)} \ll \Delta^{-j}$ give $\partial_t^j [w(t)V_{\alpha,\beta}(x,t)] \ll L^2(1+x/T)^{-A}\Delta^{-j}$ on the support, so j integrations by parts give

$$\int w(t) \left(\frac{hm}{kn}\right)^{-it} V_{\alpha,\beta}(mn,t) dt \ll HL^2 \frac{(1+mn/T)^{-A}}{(\Delta|\log(hm/kn)|)^j} \leq HL^2(1+mn/T)^{-A} \left(\frac{2\sqrt{hkmn}}{\Delta}\right)^j,$$

since $|\log(hm/kn)| \geq 1/(2\sqrt{hkmn})$ for $hm \neq kn$. Put $\eta = \theta - 1/2 - \nu > 0$ and $\epsilon = \eta/2$. If $mn \leq T^{1+\epsilon}$ then $2\sqrt{hkmn}/\Delta \leq 2yT^{(1+\epsilon)/2}L/T^\theta = 2LT^{-3\eta/4}$, and there are at most T^2 such pairs (m,n) , so these terms contribute $O(T^{-A})$ once $j > 4(A+5)/(3\eta)$. If $mn > T^{1+\epsilon}$ we do not integrate by parts: the trivial bound $HL^2(mn/T)^{-A'}$ summed against $(mn)^{-1/2+O(1/L)}$ gives $O(HL^3T^{(1+\epsilon)/2-\epsilon A'+O(1)})$, which is $O(T^{-A})$ for $A' \geq (A+3)/\epsilon$. The reflected part is identical because $t^j \partial_t^j X_{\alpha,\beta,t} \ll |X_{\alpha,\beta,t}| \ll 1$. $\square$

Inserting the mollifier, $I = I_1 + I_2 + O(T^{-A})$, where I_1 is Young's (5.1) with M replaced by y and I_2 is $I_1(-\beta, -\alpha)$ with $X_{\alpha,\beta,t}$ inserted in the t -integral. On the support of w we have $t = T(1 + O(H/T))$, so by (5) $X_{\alpha,\beta,t} = (T/2\pi)^{-\alpha-\beta}(1 + \rho(t))$ with $\rho \ll H/T$. Write $h = gh'$, $k = gk'$ with $(h', k') = 1$; then $hm = kn$ means $m = k'r$, $n = h'r$, and

$$\sum_{h,k \leq y} \frac{1}{\sqrt{hk}} \sum_{hm=kn} \frac{(1+mn/T)^{-A}}{(mn)^{1/2-O(1/L)}} \ll L \sum_{g \leq y} \frac{1}{g} \left(\sum_{h' \leq y} \frac{1}{h'} \right)^2 \ll L^4.$$

Bounding with absolute values, the contribution of ρ to I_2 is $\ll H(H/T)L^6 = o(H/L)$ because $\theta < 1$. Since $(2\pi)^{\alpha+\beta} = 1 + O(1/L)$ and $|I_1(-\beta, -\alpha)| \ll H$ by the next lemma (the region of the annuli is invariant under $(\alpha, \beta) \mapsto (-\beta, -\alpha)$),

$$I(\alpha, \beta) = I_1(\alpha, \beta) + T^{-\alpha-\beta} I_1(-\beta, -\alpha) + O(H/L). \quad (6)$$

Lemma 2.2 (Diagonal). *Uniformly on the annuli, $I_1(\alpha, \beta) = c_1(\alpha, \beta)\hat{w}(0) + O(H/L)$ with*

$$c_1(\alpha, \beta) = \frac{1}{(\alpha + \beta) \log y} \int_0^1 (P'(u) + \alpha \log y P(u))(P'(u) + \beta \log y P(u)) du.$$

Proof. Follow Young's Section 6. After the Mellin representation (6.1) and the arithmetic identity (6.2), move the u, v -contours to $\Re = \delta$ and the s -contour to $\Re s = -\delta + \epsilon_1$ with a fixed small $\epsilon_1 > 0$. The only pole crossed is $s = 0$, because G vanishes at the pole $s = -(\alpha + \beta)/2$ of $\zeta(1 + \alpha + \beta + 2s)$. The only factor involving w is $\int w(t)g_{\alpha,\beta}(s,t)dt$, and by Young's (4.2), $|g_{\alpha,\beta}(s,t)| \ll (t/2\pi)^{-\delta+\epsilon_1}(1+|s|^2)$ on the new contour for $|\Im s| \leq T^{1/3}$. This factor is therefore $\ll HT^{-\delta+\epsilon_1}(1+|s|^2)$, while $|y^{u+v}| = y^{2\delta}$; the remaining factors are as in Young. The new contour contributes $\ll HT^{-\delta(1-2\nu)+\epsilon_1}L^{O(1)} = o(H/L)$ for $\epsilon_1 < \delta(1-2\nu)/2$, since $\nu < 1/2$. The residue at $s = 0$ has $g_{\alpha,\beta}(0,t) = G(0) = 1$ and equals $\hat{w}(0)$ times the expression in Young's (6.3). Young's Lemma 7 and Section 7 involve neither w nor T and give the main term with relative error $O(1/L)$. $\square$

Proof of Theorem 1.2. As in Young's deduction of his Lemma 3 from Lemma 6,

$$c_1(\alpha, \beta) + c_1(-\beta, -\alpha) = \int_0^1 2PP' = 1, \quad \frac{1 - T^{-\alpha-\beta}}{(\alpha + \beta) \log y} = \frac{1}{\nu} \int_0^1 T^{-v(\alpha+\beta)} dv.$$

With (6) and Lemma 2.2 this gives (4) on the annuli, and the maximum modulus principle extends it to $\alpha, \beta \ll 1/L$ because both sides are holomorphic. By Young's (3.4), the left side of Theorem 1.2 equals

$$Q\left(-\frac{1}{L}\frac{\partial}{\partial\alpha}\right)Q\left(-\frac{1}{L}\frac{\partial}{\partial\beta}\right)I(\alpha, \beta)\Big|_{\alpha=\beta=-R/L};$$

the derivatives are Cauchy integrals over circles of radius $\asymp 1/L$ on which the error is uniform, and the operator applied to $c(\alpha, \beta)$ gives Young's (1.3) with ϑ replaced by ν . Since

$$\frac{d}{dx}\left[e^{R\nu x}P(x+u)Q(v+\nu x)\right]_{x=0} = e^{-Rv}(w_R(v)P'(u) + \nu w'_R(v)P(u)),$$

this is the stated constant. $\square$

Remark 2.3. The hypothesis $\nu < \theta - 1/2$ is used only in Lemma 2.1, and the bound $\nu < 1/2$ only in Lemma 2.2. The hypothesis $\theta < 1$, through $H = o(T)$, places the support of w in $[T/2, 2T]$, so that $t^{-1} \leq \Delta^{-1}$, and bounds ρ . The normalization $Q(0) = 1$ enters only through the constant term: the operator maps the term 1 of $c(\alpha, \beta)$ to $Q(0)^2$, so for every real polynomial Q the moment is $(Q(0)^2 + \nu^{-1} \int_0^1 \int_0^1 (w_R P' + \nu w'_R P^2) \widehat{w}(0) + O(H/L))$, and in particular $O(H)$.

3 Distinct sign changes

3.1 An exact identity on the critical line

Let $\chi(s) = \pi^{s-1/2} \Gamma((1-s)/2) / \Gamma(s/2)$, so that $\zeta(s) = \chi(s)\zeta(1-s)$, and put $\lambda(s) = -\chi'(s)/\chi(s)$. Write $D = d/ds$ and $Q(x) = \sum_k q_k x^k$.

Lemma 3.1. *For every $k \geq 0$, $\chi(s)\zeta^{(k)}(1-s) = (-1)^k (D + \lambda)^k \zeta(s)$, where $(D + \lambda)^k$ is the k -fold composition of $f \mapsto f' + \lambda f$. Consequently $\chi(s)V(1-s) = \sum_k q_k L^{-k} (D + \lambda)^k \zeta(s)$. If $Q(x) + Q(1-x) = \beta$ identically, then*

$$E_\beta := \beta\zeta - V - \chi(s)V(1-s) = \sum_k q_k L^{-k} [(D + L)^k - (D + \lambda)^k] \zeta(s), \quad (7)$$

and with $\widetilde{V} = V + E_\beta/2$ and $\omega(t) = e^{i\vartheta(t)}$,

$$\beta Z(t) = 2 \Re(\omega(t) \widetilde{V}(\tfrac{1}{2} + it)) \quad (t \in \mathbb{R}). \quad (8)$$

Proof. Put $f(s) = \zeta(1-s) = \chi(1-s)\zeta(s)$ and note $-\chi'(1-s)/\chi(1-s) = \lambda(s)$ because $\chi(s)\chi(1-s) = 1$. Then $f' = \chi(1-s)(D + \lambda)\zeta$ and, inductively, $f^{(k)} = \chi(1-s)(D + \lambda)^k \zeta$; since $\zeta^{(k)}(1-s) = (-1)^k f^{(k)}(s)$, the first claim follows, and the second by linearity. The polynomial identity $Q(-x) + Q(1+x) = \beta$ with $x = D/L$ gives $\beta\zeta = V + \sum_k q_k L^{-k} (D + L)^k \zeta$, which is (7). On $s = 1/2 + it$ we have $1-s = \bar{s}$, $V(\bar{s}) = \overline{V(s)}$ and $\chi(s) = \omega^{-2}$, so $\omega\chi(s)V(1-s) = \overline{\omega V(s)}$ and $\omega\zeta = Z$. Hence $\beta Z = 2\Re(\omega V) + \omega E_\beta$. The left side and $2\Re(\omega V)$ are real, so ωE_β is real, and (8) follows. $\square$

The identity (8) holds for every real Q ; the symmetry of Q makes E_β small. By Stirling's formula, $\lambda(s) = \log(|t|/2\pi) + O(1/|t|)$ and $\lambda^{(j)}(s) \ll 1/|t|$ for $j \geq 1$, uniformly in fixed vertical strips, so $\lambda - L = O(1)$ for $t \asymp T$. In normal-ordered form $(D + \lambda)^k = \sum_{m \leq k} c_{k,m} D^m$, where $c_{k,m}$

is $\binom{k}{m}\lambda^{k-m}$ plus a polynomial in $\lambda, \lambda', \dots$ of weight $k-m$ whose terms contain a derivative of λ . Hence the coefficient of D^m in $(D+L)^k - (D+\lambda)^k$ is $O(L^{k-m-1})$, and

$$E_\beta = \sum_{j < \deg Q} \varepsilon_j(s) L^{-j} \zeta^{(j)}(s), \quad \varepsilon_j(s) \ll 1/L, \quad (9)$$

uniformly for $\sigma_0 \leq \sigma \leq \sigma_1$ and $T/2 \leq t \leq 2T$. For $j \geq 1$, $L^{-j} \zeta^{(j)} = (-1)^j (V_{1+x^j} - \zeta)$, where V_{1+x^j} is the V of the polynomial $1+x^j$, so Theorem 1.2 for the polynomials 1 and $1+x^j$ and Cauchy-Schwarz give

$$\int w |\psi E_\beta(\sigma_0 + it)|^2 dt \ll H/L^2. \quad (10)$$

Published admissible polynomials often have $\beta \neq 1$; for instance Conrey's polynomial of degree seven has $\beta = 0.984$. For them the unnormalized difference $\zeta - V - \chi V(1-s)$ is not small, which is why E_β is normalized by β .

3.2 The counting lemma

Lemma 3.2. *Let $\sigma_1 = \sigma_1(P, Q)$ be fixed and large, let $L = \log T_0$ be fixed with T_0 large, and let $|T - T_0| \leq 1$, $H \asymp T_0^\theta$, where T and $T+H$ are not ordinates of zeros of Z or of $\tilde{V}$ in $[1/2, \sigma_1]$. Let $N_{\tilde{V}}$ count, with multiplicity, the zeros of $\tilde{V}$ in $[1/2, \sigma_1] \times (T, T+H]$, including those on $\sigma = 1/2$. Then*

$$O(T, H) \geq N(T, H) - 2N_{\tilde{V}} - O(L).$$

Proof. Let $t_1 < \dots < t_J$ be the zeros of $\tilde{V}(1/2 + it)$ in $(T, T+H)$, of orders $k_1, \dots, k_J$, and $K = \sum k_j$. Write $\tilde{V}(s) = \prod_j (s - 1/2 - it_j)^{k_j} F(s)$ with F holomorphic and zero-free on the segment. On the line $\omega \tilde{V} = i^K p(t) \omega F$ with $p(t) = \prod_j (t - t_j)^{k_j}$ real, so by (8) $\beta Z = 2p \Re W_0$, where $W_0 = i^K \omega F$ is continuous and zero-free on $[T, T+H]$.

Let ϕ be a continuous argument of W_0 ; assume $\phi(T) < \phi(T+H)$, the other case being symmetric, and let $c_1 < \dots < c_M$ be the points of $\pi/2 + \pi\mathbb{Z}$ in $(\phi(T), \phi(T+H))$, so that $M \geq |\Delta\phi|/\pi - 1$. Put $\tau_0 = T$, $\tau_M = T+H$, and for $0 < i < M$ let τ_i be the first point with $\phi(\tau_i) = c_i + \pi/2$. Then $\tau_0 < \dots < \tau_M$ and $\cos \phi(\tau_i)$ alternates in sign, since $\phi(\tau_0) \in (c_1 - \pi, c_1)$, $\phi(\tau_M) \in (c_M, c_M + \pi)$ and $\phi(\tau_i) = c_i + \pi/2$. After moving the interior τ_i slightly, none of them is a t_j . As $\text{sgn } Z(\tau_i) = \text{sgn}(\beta) \text{sgn } p(\tau_i) \text{sgn } \cos \phi(\tau_i)$ and p changes sign only at the t_j , the function Z has opposite signs at the ends of at least $M - J$ of the disjoint intervals (τ_{i-1}, τ_i) , and each such interval contains a sign change. Hence $O(T, H) \geq |\Delta\phi|/\pi - 1 - J$.

Apply the argument principle to F on $[1/2, \sigma_1] \times [T, T+H]$. It is zero-free on the left side, and its zeros inside are the $N_{>}$ zeros of $\tilde{V}$ with $\sigma > 1/2$. On the bottom, right and top sides $\arg F = \arg \tilde{V} - \sum_j k_j \arg(s - 1/2 - it_j)$, and each factor turns by exactly $+\pi$ along these sides. Therefore

$$\Delta\phi = \Delta\vartheta + \Delta_{\text{brt}} \arg \tilde{V} - \pi K - 2\pi N_{>}.$$

The Riemann-von Mangoldt formula gives $\Delta\vartheta/\pi = N(T, H) + O(L)$. On $\sigma = \sigma_1$ we have $|V - 1| \leq 1/4$ and $|E_\beta| \ll 1/L$, so $\Re \tilde{V} > 0$ and that side contributes $O(1)$. On the horizontal sides the Backlund-Jensen argument gives $O(L)$, because $\tilde{V} = (\beta\zeta + V - \chi V(1-s))/2 = \sum_{j \leq \deg Q} a_j(s) \zeta^{(j)}(s)$ with each a_j a polynomial in L^{-1} , λ and its derivatives, so $\tilde{V}$ is holomorphic and polynomially bounded near the window and $\Re \tilde{V} \geq 1/2$ at $\sigma_1 + iT$. Since $J \leq K$ and $N_{\tilde{V}} = N_{>} + K$,

$$O(T, H) \geq N(T, H) - 2(N_{>} + K) - O(L) = N(T, H) - 2N_{\tilde{V}} - O(L). \quad \square$$

For a general window we keep $L = \log T$ and move both endpoints inward by less than one unit to avoid the finitely many exceptional ordinates, including those of zeros of $\psi\tilde{V}$; each unit interval contains $O(L)$ zeros of ζ and, by Jensen's formula, $O(L)$ zeros of $\tilde{V}$ and of $\psi\tilde{V}$ in $[\sigma_0, \sigma_1]$, so every count changes by $O(L)$. The quantity J in the proof may be replaced by the number of t_j of odd order.

A double zero of Z is not a sign change and is never counted; it is paid for in $N_{\tilde{V}}$, as an on-line zero of $\tilde{V}$ when $\deg Q = 1$ and, in the examples of Section 5, as a zero of $\tilde{V}$ to the right of the line when $\deg Q = 3$. If the on-line zeros of $\tilde{V}$ are omitted from $N_{\tilde{V}}$, the inequality fails (Section 5).

3.3 Proof of Theorem 1.3

Let $F_1 = \psi\tilde{V}$. Littlewood's lemma on $[\sigma_0, \sigma_1] \times [T, T+H]$ gives

$$2\pi \int_{\sigma_0}^{\sigma_1} n(\sigma) d\sigma = \int_T^{T+H} \log |F_1(\sigma_0 + it)| dt - \int_T^{T+H} \log |F_1(\sigma_1 + it)| dt + O(L),$$

where $n(\sigma)$ counts the zeros of F_1 in the rectangle with real part greater than σ , and $O(L)$ bounds the horizontal argument integrals: ψ and $\tilde{V}$ are polynomially bounded and have real part at least $1/2$ at $\sigma_1 + iT$ and $\sigma_1 + i(T+H)$, so the Backlund–Jensen argument bounds the variation of $\arg \psi$ and of $\arg \tilde{V}$, hence of $\arg F_1$. Each zero counted in $N_{\tilde{V}}$ has real part at least $1/2 = \sigma_0 + R/L$, and every other zero contributes nonnegatively, so the left side is at least $2\pi(R/L)N_{\tilde{V}}$.

On $\sigma = \sigma_1$, $\psi V = 1 + \sum_{n \geq 2} b_n n^{-s}$ with $|b_n| \ll d(n)(\log n)^{\deg Q}$, so its logarithm is an absolutely convergent Dirichlet series for σ_1 large and $\int \log |\psi V(\sigma_1 + it)| dt = O(1)$; moreover $|E_\beta/(2V)| \ll 1/L$ there, which contributes $O(H/L)$. On $\sigma = \sigma_0$, concavity of the logarithm, $w = 1$ on $[T, T+H]$ with $w \geq 0$, the inequality $|a+b|^2 \leq (1+L^{-1/2})|a|^2 + (1+L^{1/2})|b|^2$, Theorem 1.2 and (10) give

$$\int_T^{T+H} \log |F_1(\sigma_0 + it)| dt \leq \frac{H}{2} \log \left(\frac{1}{H} \int w |\psi \tilde{V}|^2 \right) \leq \frac{H}{2} \log c(P, Q, R, \nu) + O(HL^{-1/2}).$$

Hence $N_{\tilde{V}} \leq (HL/4\pi R) \log c + o(HL)$. Since $N(T, H) = (HL/2\pi)(1 + o(1))$, Lemma 3.2 gives $O/N \geq 1 - R^{-1} \log c - o(1)$. $\square$

Remark 3.3. The number σ_1 and all implied constants depend on P, Q, R and ν . For the polynomials of Section 5 the monomial coefficients of Q have size about 10^{120} and R is about 28, so the constants are very large, and for $\eta = \theta - 1/2 - \nu = 10^{-4}$ the factor $2LT^{-3\eta/4}$ of Lemma 2.1 is below one only when $\log T$ exceeds about $1.7 \cdot 10^5$. The results are purely asymptotic.

4 Transfer to simple zeros

Theorem 4.1 (D). *If $\liminf O(T, T^\theta)/N(T, T^\theta) \geq k \geq 0$, then $\liminf S(T, T^\theta)/N(T, T^\theta) \geq h(\theta; k)$.*

Proof. This is the argument of [2, proof of Theorem 1.2] with the odd-support input replaced by k . For a real even $\eta \in C_c^\infty((-\lambda/2, \lambda/2))$, $0 < \lambda < \theta$, with $\int \eta^2 = 1$, let Q_T be the pair sum of $f = \eta^2$ over the scaled zeros of the window; it is unrelated to the polynomial Q of the previous sections. With q_T, s_T, o_T the ratios of Q_T, S, O to N , the Hilbert–parity inequality [2, Proposition 2.5] reads $(q_T - s_T)(1 - o_T) \geq 2(1 - s_T)^2$, and Wang's theorem [10, Theorem 2.2] gives $q_T \rightarrow \mathcal{C}(f)$. If $O = N$, every zero of the window is simple and critical and $s_T = 1$. Otherwise $1 - o_T > 0$ and the inequality gives $q_T \geq s_T$; for every $\epsilon > 0$ and all large T , $q_T \leq \mathcal{C}(f) + \epsilon$ and

$o_T \geq k - \epsilon$, and enlarging the nonnegative factors gives $2(1 - s_T)^2 \leq (\mathcal{C}(f) + \epsilon - s_T)(1 - k + \epsilon)$. Hence $\liminf s_T \geq R(\mathcal{C}(f), 1 - k)$ with $R(A, b) = (4 - b - \sqrt{b(b + 8(A - 1))})/4$, the smaller root. Letting f tend to the cosine density at fixed λ , so that $\mathcal{C}(f) \rightarrow 2 - c(\lambda)$, and then $\lambda \rightarrow \theta$, gives $R(2 - c(\theta), 1 - k) = h(\theta; k)$. $\square$

For fixed $k < 1$, $h(\theta; k)$ increases with c and with k , since $\partial_k[(1 - k)(9 - k - 8c)] = -(10 - 2k - 8c) < 0$ for $c < 1$. As $c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) > 0$, if $h(\theta_*; \kappa) > 0$ for a detector with $\nu < \theta_* - 1/2$, then the same detector is admissible and $h(\theta; \kappa) > 0$ for every $\theta \in [\theta_*, 1)$. The linear branch $(c + 2k)/3$ in (3) is the transfer of [2, Theorem 1.4] with the same input.

5 Certified detector constants

We use Q in the Chebyshev family

$$Q(y) = 1 - y + \sum_{j=1}^{100} x_j (T_{2j+1}(1 - 2y) - (1 - 2y)),$$

which has degree 201, $Q(0) = 1$, $Q(1) = 0$ and $Q(y) + Q(1 - y) = 1$ for all rational x_j , because T_{2j+1} is odd with $T_{2j+1}(1) = 1$, and $P(x) = S(rx)/S(r)$ with S the Taylor polynomial of $\sinh$ of degree 13. The coefficients x_j are rationals with denominator 10^{40} obtained by minimizing c numerically, and R, r, ν are rational. With $B_P = \int_0^1 P^2$, $C_P = \int_0^1 P'^2$ and $\int_0^1 PP' = 1/2$, and since $\int_0^1 w_R w'_R = (e^{2R}Q(1)^2 - Q(0)^2)/2 = -1/2$,

$$c(P, Q, R, \nu) = \frac{1}{2} + \frac{C_P}{\nu} \int_0^1 e^{2Rv} Q(v)^2 dv + \nu B_P \int_0^1 e^{2Rv} (RQ(v) + Q'(v))^2 dv.$$

Each integral $\int_0^1 e^{2Rv} F(v) dv$ of a rational polynomial equals $e^{2R}a(F) + b(F)$ with rational a, b , by the recursion $\int_0^1 e^{tv} v^k dv = e^t a_k + b_k$, $a_k = 1/t - (k/t)a_{k-1}$, $b_k = -(k/t)b_{k-1}$, $a_0 = 1/t$, $b_0 = -1/t$. Hence $c = Ae^{2R} + B$ with exact rationals A, B , and $\kappa = 1 - R^{-1} \log c$ needs one exponential and one logarithm in ball arithmetic (Arb, 4000 bits). The admissibility identities are verified in exact rational arithmetic.

Table 1: Certified detector constants. Lower bounds are rounded down.

ν	R	r	$c(P, Q, R, \nu)$	$\kappa >$	$\kappa/\nu >$
0.0199	48.391	0.7702	$5.1998 \cdot 10^{20}$	0.0142729911	0.717235
0.0299	32.244	0.7708	$5.0471 \cdot 10^{13}$	0.0214483010	0.717334
0.0339	28.439	0.7708	$1.1235 \cdot 10^{12}$	0.0243176419	0.717334
0.0349	27.625	0.7709	$4.9777 \cdot 10^{11}$	0.0250349766	0.717334
0.0399	24.163	0.7709	$1.5614 \cdot 10^{10}$	0.0286216496	0.717334
0.0458	21.05	0.7708	$6.9431 \cdot 10^8$	0.0328539236	0.717334
0.0499	19.321	0.7709	$1.2321 \cdot 10^8$	0.0357949954	0.717334
0.0999	9.648	0.7707	$7.7588 \cdot 10^3$	0.0716640051	0.717357

In our numerical optimization the ratio κ/ν did not exceed about 0.7173 for this mollifier shape at small ν ; the polynomials of Conrey, Farmer, Kwan, Lin and Turnage-Butterbaugh [3] give about 0.682ν . Linear Q does not suffice at these lengths: $Q = 1 - x$, $P = x$ needs $\nu > 0.1928$

for $\kappa > 0$. The same code reproduces Young's $c = 2.35006777\dots$ for $P = x$, $Q = 1 - x$, $R = 1.3$, $\nu = 1/2$, and Conrey's $\kappa > 0.4088$ for his polynomial of degree seven with $\nu = 4/7$.

An independent program evaluates Q by the Chebyshev recurrence directly from the x_j and P by its series, separates the double integral by Fubini into one-dimensional integrals, computes each by validated Arb quadrature, and recomputes $c(\theta)$ and h with 100-digit interval arithmetic in `mpmath`. Every enclosure overlaps the primary one, and all thresholds hold again. With $\nu = \theta - 1/2 - 10^{-4}$, the directed values of $h(\theta; \kappa)$ at $\theta = 0.534, 0.535, 0.54, 0.5459, 0.55$ exceed $1.4806994 \cdot 10^{-5}$, 0.0015246940 , 0.0090231376 , 0.0177638490 and 0.0237708528 ; at $\theta = 0.5459$ the ratio to the rank-six Selberg value $1.7764518 \cdot 10^{-5}$ exceeds 999.96.

Numerical controls support Lemma 3.2. At $T = 10^4, 10^5, 10^6$ and for Q of degrees 1, 3, 5, 7, including Conrey's polynomial and polynomials with $\beta \neq 1$, the direct evaluation of $\chi(s)V(1-s)$ agrees with Lemma 3.1, ωE_β is real and (8) holds to the working precision. On windows at these heights the sign changes of Z , the level crossings of $\arg W_0$ and the zero counts coincide. For the test functions $f = \zeta \prod_j ((s - 1/2)^2 + \gamma_j^2)^{e_j}$ with planted double ($e_j = 1$) and triple ($e_j = 2$) critical zeros, the inequality of Lemma 3.2 holds with equality in the double-zero cases, double zeros are not counted, and the variant that omits the on-line zeros of $\tilde{V}$ fails for $\deg Q = 1$.

6 Scope

Theorems 1.2 and 1.3 are unconditional and use only classical tools together with Young's argument. Theorem 1.1 also uses Wang's short-interval pair theorem [10], a recent preprint, through the finite inequality of [2]. All statements are asymptotic and give no effective starting height. They do not show that every zero in the window is simple or on the critical line, and they do not prove the Riemann hypothesis.

The onset is limited by the mollifier range $\nu < \theta - 1/2$, which is the classical range; with linear Q it gives only $\theta \geq 0.693$. The gain comes from operator polynomials of high degree, which lose the simple-zero information, together with the parity inequality, which restores it. A mollified moment with general Q in the range $\nu < (3\theta - 1)/4$, which Steuding established for $Q(x) = 1 - x$ [9], would, by a numerical evaluation with slopes as in Table 1, make the positivity condition $\kappa > 1 - 2/(2 - c(\theta))$ hold for every $\theta > 1/2$; we do not prove such a moment here.

Data availability. The programs, exact parameters, outputs and independent replays are archived as computation EXP-010 in the repository [7].

Declaration of generative AI use. Generative AI tools were used for source research, derivation, review and implementation. The author reviewed and verified all content and takes full responsibility for the work.

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